

Chill. It's just a few pieces of paper. Each question is worth 2 points except where noted.

Match the *most appropriate* term to each concept. Not all terms are used. No term is used twice.

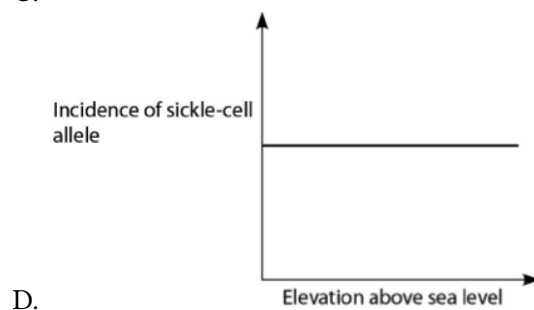
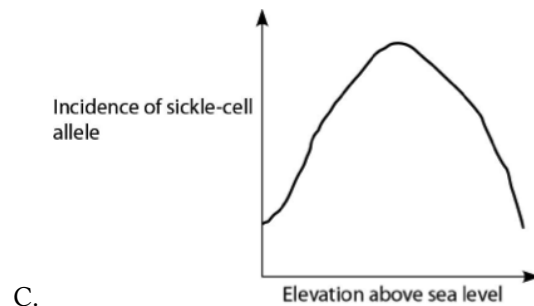
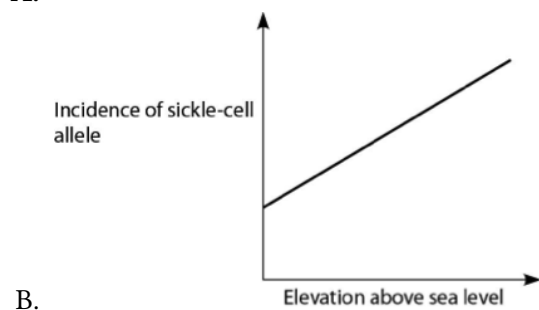
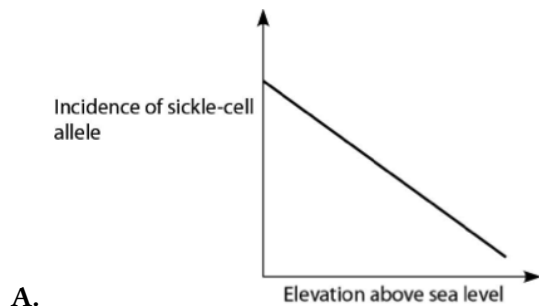
adaptation	environment	Thomas Robert Malthus
allelotype	genotype	mutation
artificial selection	haploid	natural selection
balancing selection	independent evolution	phenotype
convergent evolution	inductive	population selection
Georges Cuvier	Jean Baptiste de Lamarck	scientific
deductive	Carolus Linnaeus	stabilizing selection
diploid	Charles Lyell	Alfred Russel Wallace

1. **Diploid** Has two alleles at a gene.
2. **Genotype** The combination of alleles at a gene.
3. **Balancing Selection** Can cause stabilizing selection in a population.
4. **Inductive** Reasoning that uses specific observations to develop general principles.
5. **Convergent evolution** Unrelated organisms evolve similar traits by adapting to similar habitats.
6. **Mutation** Creates new genetic variation within a population.
7. **Environment** Contributes to the phenotype but is not inherited.
8. **Adaptation** Trait that increases fitness in the current environment.
9. **Lyell** Argued that Earth changed uniformly for much more than 6000 years.
10. **Lamarck** Was the first to propose that organisms change over time.
11. (4 points) Briefly compare sexual selection to natural selection to explain why sexual selection can cause evolutionary change more rapidly than other forms of natural selection.

Sexual selection acts directly on reproductive success while natural selection is indirect.

Multiple choice. Choose the *best* answer. Clearly circle the letter for your answer. Vague or unclear marks will be counted as incorrect.

12. The rabbitsfoot mussel was once widespread throughout rivers of eastern North America but now remains in only a few very small populations. The decrease occurred rapidly due to habitat loss. Genetics studies show that the remaining populations have little or no genetic variation. The best explanation for this result is:
- A. Inbreeding caused an increase in heterozygosity.
 - B. Balancing selection favored only one or two alleles.
 - C. Genetic drift caused the loss of heterozygosity.**
 - D. A bottleneck caused the loss of homozygosity.
 - E. Directional selection favored only one or two alleles.
13. Which *one* of the following statements is true of natural selection?
- A. It requires genetic variation, results in descent with modification, and involves differences in reproductive success.**
 - B. It requires genetic variation.
 - C. It results in descent with modification.
 - D. It involves differences in reproductive success.
 - E. It results in descent with modification and involves differences in reproductive success.
14. *Anopheles* mosquitoes, which carry the malaria parasite, cannot live above elevations of 5,900 feet. In addition, oxygen availability decreases with higher altitude. Consider a hypothetical human population that is adapted to life on the slopes of Mt. Kilimanjaro in Tanzania, a country in equatorial Africa. Mt. Kilimanjaro's base is about 2,600 feet above sea level and its peak is 19,341 feet above sea level. If the frequency of the sickle-cell allele in the population is plotted against altitude (feet above sea level), which of the following distributions is most likely, assuming little migration of people up or down the mountain?



15. Which mouse has the greatest relative fitness? (A clutch is the offspring from one breeding event.)
- A. One that lives for 2 years, has 3 clutches per year, and 8 offspring per clutch.
 - B. One that lives for 4 years, has 3 clutches per year, and 4 offspring per clutch.
 - C. One that lives for 3 years, has 3 clutches per year, and 5 offspring per clutch.
 - D. One that lives for 3 years, has 3 clutches per year, and 7 offspring per clutch.**
 - E. One that lives for 3 years, has 2 clutches per year, and 6 offspring per clutch.
16. Organisms living in the northern rocky intertidal face competing selective forces. Larger individuals benefit by having a larger internal volume, which helps to conserve body temperature. However, smaller individuals are less likely to be dislodged by large waves from storms. Thus, selection may favor intermediate body size phenotype for any particular species. The mode of selection is
- A. natural selection.
 - B. phenotype selection.
 - C. disruptive selection.
 - D. stabilizing selection.**
 - E. directional selection.
17. In science, a hypothesis is only useful if it
- A. agrees with the conclusion.
 - B. is proven correct.
 - C. can be tested.**
 - D. can be proven correct.
 - E. explains the natural world.
18. The overuse of drugs such as penicillin has caused an increase in the number of bacterial species that are resistant to the drugs. The most likely explanation for this result is:
- A. Bacteria began making proteins resistant to the drugs.
 - B. A few bacteria were naturally resistant and natural selection favored their increase.**
 - C. Most bacteria can resist drugs.
 - D. The drugs caused mutations in the bacteria that gave them resistance.
 - E. Penicillin was not an effective drug.
19. According to Darwin's concept of descent with modification:
- A. All organisms overproduce offspring, leading to differences in survivorship.
 - B. All species have evolved from a single common ancestor.
 - C. All organisms overproduce offspring, leading to different adaptations.
 - D. Individuals with greater fitness will evolve more quickly than individuals with lower fitness.
 - E. Species evolve from ancestral forms through the accumulation of different adaptations.**
 - F. Populations evolve by natural selection acting on individuals.
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20. Which of these conditions should completely prevent the occurrence of natural selection in a population over time?
- A. The population size is large.
 - B. The population lives in a habitat where there are no competing species present.
 - C. The environment is changing at a relatively slow rate.
 - D. All variation between individuals is due only to environmental factors.**
 - E. The population mates non-randomly.
21. Violation of these two Hardy-Weinberg assumptions always causes homozygosity to increase.
- A. Natural selection and gene flow.
 - B. Genetic drift and natural selection.
 - C. Non-random mating and gene flow.
 - D. Genetic drift and non-random mating.**
 - E. Mutation and natural selection.
 - F. Gene flow and mutation.
22. (5 points) Here is the full classification of the Scalloped Hammerhead Shark.
- Eukaryota; Metazoa; Chordata; Chondrichthys; Carcharhiniformes; Sphyrnidae; Sphyrna; lewini
- (a) Which is the order? **Carcharhiniformes**
- (b) Write the scientific name using proper binomial nomenclature. ***Sphyrna lewini***

Circle True or False. Read each statement completely before answering. Take each statement at face value.

- 23. **TRUE** False Genetic drift is strongest in small populations.
 - 24. **TRUE** False Adaptations can lead to descent with modification.
 - 25. **TRUE** False The smallest unit of evolution is the population.
 - 26. True **FALSE** Overproduction of offspring by every individual in a population is necessary for evolution to occur.
 - 27. True **FALSE** The genotype is determined by the interaction of the phenotype with the environment.
 - 28. True **FALSE** One assumption of the Hardy-Weinberg equations is that mutations are random.
 - 29. True **FALSE** Thomas Robert Malthus wrote *Systema Naturæ* that influenced Darwin.
 - 30. **TRUE** False If $p = 0.4$, then total homozygosity in the population is 0.52 (assume equilibrium).
 - 31. **TRUE** False Sexual selection usually causes a pattern of directional selection.
 - 32. **TRUE** False Genetic drift always increases homozygosity in a population.
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33. (16 points) List below the four missing steps of the Scientific Method. Then, for each step, write a statement that illustrates how scientists applied the scientific method to the experiments described below. I have provided step four as an example.

Houseflies are infamous for walking all over food around the house and at picnics. They stick out their proboscis (the fly version of a tongue) when they walk on food but not when they walk on other things, like the table. Scientists thought that flies might be able to taste the food with their feet, and stick out their proboscis whenever they identified something tasty. To test this idea, scientists filled one saucer with sugar water (sugar is a food source for the fly) and one saucer with plain water. They then glued live flies to the end of a slender stick. The flies were gently lowered until their feet just touched the liquid. If the liquid was plain water, the flies did not stick out their proboscis. If the liquid was sugar water, the flies always stuck out their proboscis.

(a) **Observation:** Flies stick out their tongue when walking on food.

(b) **Hypothesis:** Flies taste with their feet.

(c) **Prediction:** If fly feet are touched to a food source, they will stick out their proboscis.

(d) **Experiment and results:** Scientists noted that flies stuck out their proboscis when their feet touched sugar water but not when their feet touched plain water.

(e) **Conclusion:** The hypothesis was supported.

2 points for each step and 2 points for each explanation.

You may use a calculator to solve the following problem but you must **show your work** in the space provided. You may **round to 3 decimal places after each step or after the final calculation**. Any other rounding will result in a 3 point deduction.

34. (15 points) Impatiens is a type of flower that has three flower colors. Red color is produced by RR individuals. Pink color is produced by Rr individuals. White color is produced by rr individuals. A random sample of 263 flowers found that 181 were red, 72 were pink, and 10 were white. Calculate the two allele frequencies and the three genotype frequencies. *Show whether the population is in Hardy-Weinberg equilibrium.*

The final values may sum to 0.999, 1.000 or 1.001 due to rounding error. (2 points each plus 5 points for correct work.)

	Frequency
R	0.825
r	0.175
RR	0.688
Rr	0.274
rr	0.038

Be sure to show whether the population is in Hardy-Weinberg equilibrium and *show all work*.

Sums to 1.0.